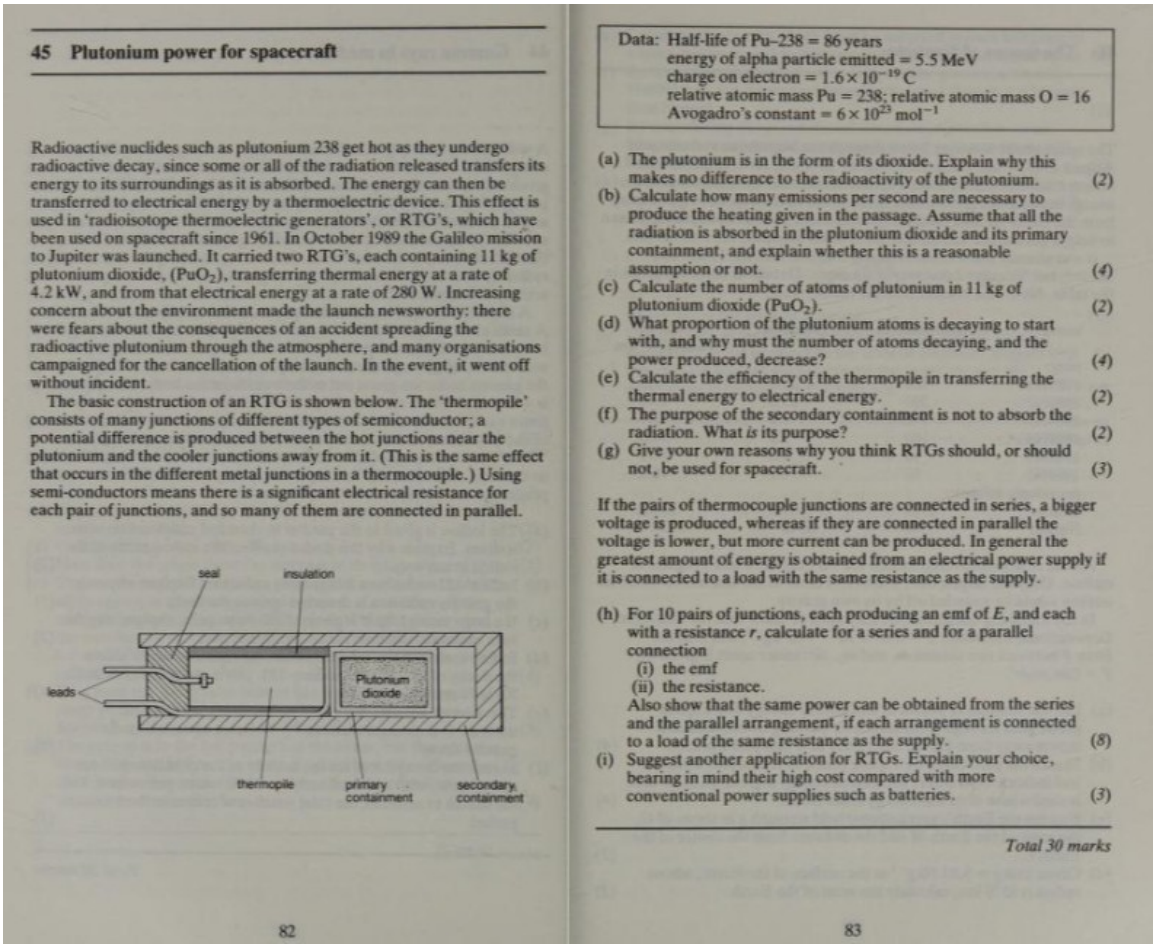
CIE 9702 Paper 4 Style (Harder): Plutonium RTG for Deep-Space Probe

A deep-space probe is powered by a radioisotope thermoelectric generator (RTG) that uses 11.0 kg of plutonium-238 dioxide, PuO_2, sealed in a robust primary container inside secondary shielding. The isotope ^{238}Pu has a half-life of 86.0 years and emits α-particles. Each decay releases about 5.50 MeV, which is absorbed in the fuel and container as thermal energy. Thermocouple junctions convert part of this thermal power into electrical power for the probe instruments. At launch, the electrical output is measured as 2.20 kW.

A simplified cross-section of the RTG is shown. The original source diagram is also provided for reference.



45 Plutonium power for spacecraft

Radioactive nuclides such as plutonium 238 get hot as they undergo radioactive decay, since some or all of the radiation released transfers its energy to its surroundings as it is absorbed. The energy can then be transferred to electrical energy by a thermoelectric device. This effect is used in 'radioisotope thermoelectric generators', or RTG's, which have been used on spacecraft since 1961. In October 1989 the Galileo mission to Jupiter was launched. It carried two RTG's, each containing 11 kg of plutonium dioxide, (PuO_2), transferring thermal energy at a rate of 4.2 kW, and from that electrical energy at a rate of 280 W. Increasing concern about the environment made the launch newsworthy: there were fears about the consequences of an accident spreading the radioactive plutonium through the atmosphere, and many organisations campaigned for the cancellation of the launch. In the event, it went off without incident.

The basic construction of an RTG is shown below. The 'thermopile' consists of many junctions of different types of semiconductor; a potential difference is produced between the hot junctions near the plutonium and the cooler junctions away from it. (This is the same effect that occurs in the different metal junctions in a thermocouple.) Using semi-conductors means there is a significant electrical resistance for each pair of junctions, and so many of them are connected in parallel.

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Data: Half-life of Pu–238 = 86 years
energy of alpha particle emitted = 5.5 MeV
charge on electron = 1.6×10^{-19} C
relative atomic mass Pu = 238; relative atomic mass O = 16
Avogadro's constant = 6×10^{23} mol^{-1}

(a) The plutonium is in the form of its dioxide. Explain why this makes no difference to the radioactivity of the plutonium. (2)
(b) Calculate how many emissions per second are necessary to produce the heating given in the passage. Assume that all the radiation is absorbed in the plutonium dioxide and its primary containment, and explain whether this is a reasonable assumption or not. (4)
(c) Calculate the number of atoms of plutonium in 11 kg of plutonium dioxide (PuO_2). (2)
(d) What proportion of the plutonium atoms is decaying to start with, and why must the number of atoms decaying, and the power produced, decrease? (4)
(e) Calculate the efficiency of the thermopile in transferring the thermal energy to electrical energy. (2)
(f) The purpose of the secondary containment is not to absorb the radiation. What *is* its purpose? (2)
(g) Give your own reasons why you think RTGs should, or should not, be used for spacecraft. (3)

If the pairs of thermocouple junctions are connected in series, a bigger voltage is produced, whereas if they are connected in parallel the voltage is lower, but more current can be produced. In general the greatest amount of energy is obtained from an electrical power supply if it is connected to a load with the same resistance as the supply.

(h) For 10 pairs of junctions, each producing an emf of E, and each with a resistance r, calculate for a series and for a parallel connection
(i) the emf
(ii) the resistance.
Also show that the same power can be obtained from the series and the parallel arrangement, if each arrangement is connected to a load of the same resistance as the supply. (8)
(i) Suggest another application for RTGs. Explain your choice, bearing in mind their high cost compared with more conventional power supplies such as batteries. (3)

Total 30 marks

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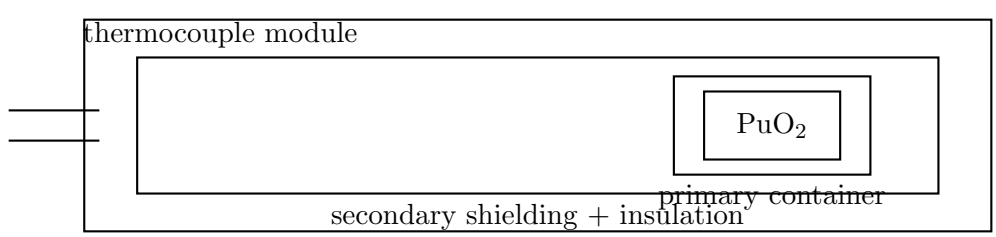

For calculations, use:

- electron charge: $e = 1.60 \times 10^{-19}\,\mathrm{C}$
- Avogadro constant: $N_A = 6.02 \times 10^{23}\,\mathrm{mol}^{-1}$
- relative atomic masses: Pu = 238, O = 16
- $1\,\mathrm{year} = 3.15 \times 10^{7}\,\mathrm{s}$

Assume each ^{238}Pu nucleus that decays releases one α-particle and that all α energy is absorbed as thermal energy inside the primary container.

(a) Explain why the fuel is used as PuO_2 pellets in sealed containers rather than as pure metallic plutonium. Your answer should refer to both chemical and radiological safety. [3]

(b) Show that the number of moles of PuO_2 in 11.0 kg is approximately 4.07×10^{1} mol. [2]

(c) Determine the number of ^{238}Pu nuclei initially present. [2]

(d)

(i) Calculate the decay constant λ in s^{-1}. [2]

(ii) Calculate the initial activity A_0 of the source. [2]

(e) Hence calculate the initial thermal power output $P_{\mathrm{th},0}$ of the fuel in watts. [3]

(f)

(i) Calculate the overall conversion efficiency from thermal to electrical power at launch. [2]

(ii) State two physical reasons why this efficiency is much less than 100%. [2]

(g) The probe requires at least 1.50 kW electrical power for full science operation. Assume the thermoelectric conversion efficiency remains constant with time.

(i) Derive an expression for electrical power $P_e(t)$ at time t after launch. [2]

(ii) Estimate the maximum mission duration for full science operation. [3]

(h) The thermocouple assembly can be modeled as n identical junction units. Each unit has emf E and internal resistance r.

(i) Write expressions for total emf and total internal resistance for:

- all n units connected in series,
- all n units connected in parallel.

[2]

(ii) For each arrangement, show that the maximum power delivered to an external load is the same, and find this maximum in terms of n, E, and r. [4]

(iii) In practice, RTGs often use mixed series-parallel networks. Suggest one engineering reason for this choice. [1]

(i) During atmospheric launch failure, one concern is dispersal of radioactive particles. State and explain two design features of the RTG that reduce environmental contamination risk in such an event. [2]

(j) A mission designer proposes replacing ^{238}Pu with a fuel of much shorter half-life but larger decay energy per nucleus. Discuss one advantage and one disadvantage of this proposal for long-duration deep-space missions. [2]

Total: 34 marks

Suggested Solution (Concise)

(a) PuO_2 is chemically more stable and less reactive than metallic Pu, so it is safer at high temperature. Ceramic pellets and sealed containment reduce dispersal risk; radiation hazard is mainly internal if particles are inhaled/ingested.

(b)

$$M(PuO_2) = 238 + 2(16) = 270\,\mathrm{g\,mol^{-1}}$$

$$n = \frac{11.0 \times 10^3}{270} = 40.7\,\mathrm{mol} = 4.07 \times 10^1\,\mathrm{mol}$$

(c) Each mole of PuO_2 contains one mole of Pu atoms.

$$N_0 = nN_A = 40.7(6.02 \times 10^{23}) \approx 2.45 \times 10^{25}\ \text{nuclei}$$

(d)(i)

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{86.0 \times 3.15 \times 10^7} = 2.56 \times 10^{-10}\,\mathrm{s^{-1}}$$

(d)(ii)

$$A_0 = \lambda N_0 = (2.56 \times 10^{-10})(2.45 \times 10^{25}) \approx 6.27 \times 10^{15}\,\mathrm{Bq}$$

(e) Energy per decay:

$$E_d = 5.50 \times 10^6 \times 1.60 \times 10^{-19} = 8.80 \times 10^{-13}\,\mathrm{J}$$

Thermal power:

$$P_{\mathrm{th},0} = A_0 E_d = (6.27 \times 10^{15})(8.80 \times 10^{-13}) \approx 5.52 \times 10^3\,\mathrm{W}$$

(f)(i)

$$\eta = \frac{P_e}{P_{\mathrm{th},0}} = \frac{2.20 \times 10^3}{5.52 \times 10^3} \approx 0.399 \approx 40\%$$

(f)(ii) Any two: limited thermoelectric figure of merit, unavoidable heat leakage, finite temperature difference, internal electrical resistance losses.

(g)(i) Since activity (and thermal power) decays exponentially,

$$P_e(t) = P_{e0}e^{-\lambda t}, \quad P_{e0} = 2.20 \times 10^3\,\mathrm{W}$$

(g)(ii)

$$1.50 \times 10^3 = 2.20 \times 10^3 e^{-\lambda t} \Rightarrow t = \frac{1}{\lambda}\ln\left(\frac{2.20}{1.50}\right)$$

$$t = \frac{\ln(1.4667)}{2.56 \times 10^{-10}} \approx 1.49 \times 10^9\,\mathrm{s} \approx 47\,\text{years}$$

(h)(i) Series: total emf $= nE$, total resistance $= nr$. Parallel: total emf $= E$, total resistance $= r/n$.

(h)(ii) Maximum load power occurs when $R_L = R_{\text{int}}$, so

$$P_{\text{max}} = \frac{\mathcal{E}^2}{4R_{\text{int}}}$$

Series:

$$P_{\text{max},s} = \frac{(nE)^2}{4nr} = \frac{nE^2}{4r}$$

Parallel:

$$P_{\text{max},p} = \frac{E^2}{4(r/n)} = \frac{nE^2}{4r}$$

Hence both are equal.

(h)(iii) Mixed networks allow required bus voltage while keeping current, heating, and reliability within design limits.

(i) Any two explained: ceramic fuel form (reduced aerosolization), robust primary capsule, secondary impact/heat shielding, multiple containment barriers.

(j) Advantage: higher initial power density (more power per unit mass initially). Disadvantage: power decays faster, so poor long-term output for multi-decade missions.